

EX NO:2	Chebyshev IIR Filter
DATE:	2.1. Design and Performance Evaluation of a Chebyshev Type-I IIR Low-Pass Filter using MATLAB

Code:

```

clc;
clear;
close all;

%-----
% EX NO : 2
% TITLE : Design and Performance Evaluation of a
%         Chebyshev Type-I IIR Low-Pass Filter
%         using MATLAB
%-----

% Filter Specifications
Wp = 0.40;    % Passband Edge Frequency (Normalized)
Ws = 0.55;    % Stopband Edge Frequency (Normalized)
Rp = 1;       % Passband Ripple (dB)
As = 40;      % Stopband Attenuation (dB)

% Determine Minimum Filter Order
[N, Wn] = cheb1ord(Wp, Ws, Rp, As);

% Design Chebyshev Type-I Low-Pass Filter
[b, a] = cheby1(N, Rp, Wn, 'low');

%-----
% Check Stability
%-----
if all(abs(roots(a)) < 1)
    stability = 'STABLE';
else
    stability = 'UNSTABLE';
end

%-----
% Frequency Response
%-----
figure;
freqz(b, a, 1024);
title('Frequency Response of Chebyshev Type-I Low-Pass Filter');

%-----
% Impulse Response

```

```

%-----
figure;
impz(b, a);
title('Impulse Response');

%-----
% Group Delay
%-----
figure;
grpdelay(b, a);
title('Group Delay');

%-----
% Pole-Zero Plot
%-----
figure;
zplane(b, a);
title('Pole-Zero Plot');

%-----
% Display Results
%-----
fprintf('-----\n');
fprintf('Chebyshev Type-I Low-Pass Filter\n');
fprintf('-----\n');
fprintf('Filter Order (N) = %d\n', N);
fprintf('Normalized Cutoff Frequency (Wn) = %.4f\n', Wn);
fprintf('Passband Edge Frequency (Wp) = %.2f\n', Wp);
fprintf('Stopband Edge Frequency (Ws) = %.2f\n', Ws);
fprintf('Passband Ripple = %d dB\n', Rp);
fprintf('Stopband Attenuation = %d dB\n', As);
fprintf('System Stability = %s\n', stability);
fprintf('-----\n');

```

Output:

Fig 1:

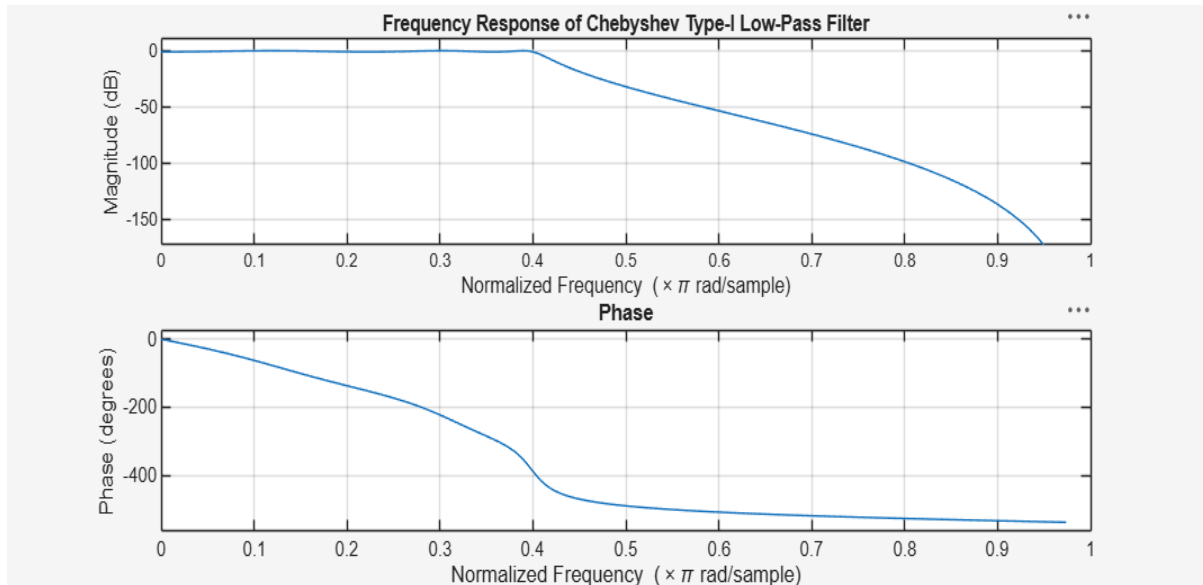


Fig 2:

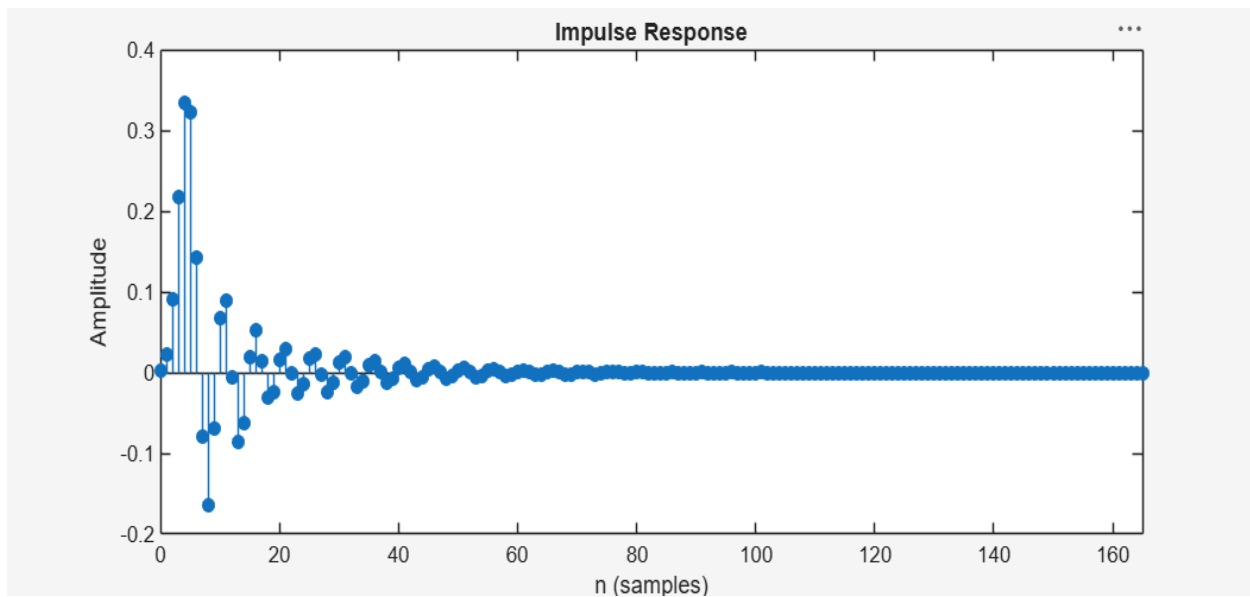


Fig 3:

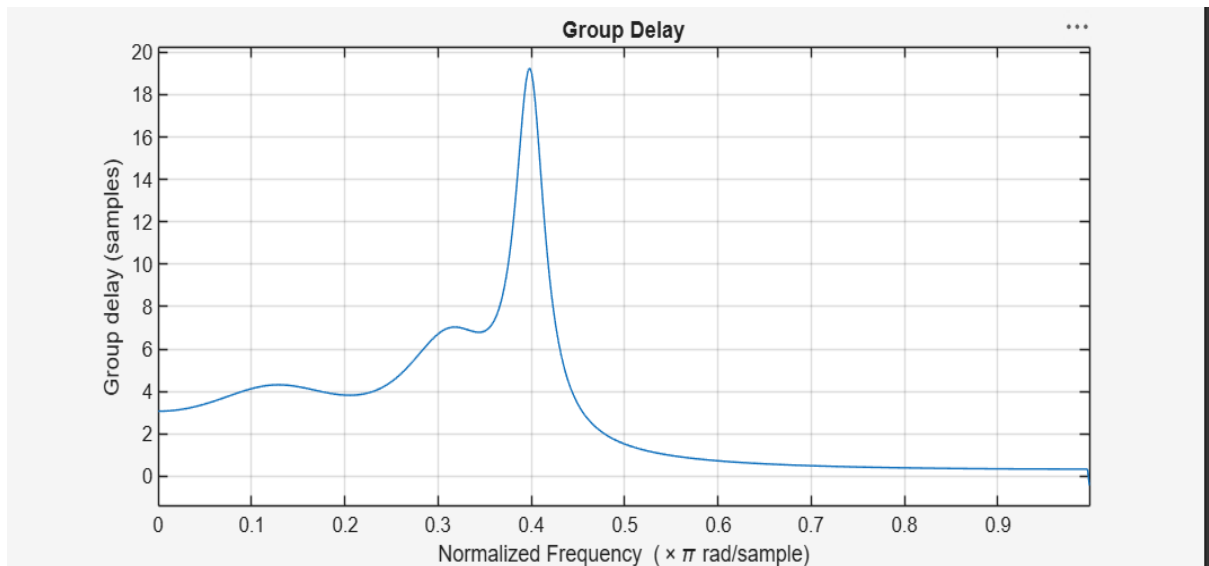
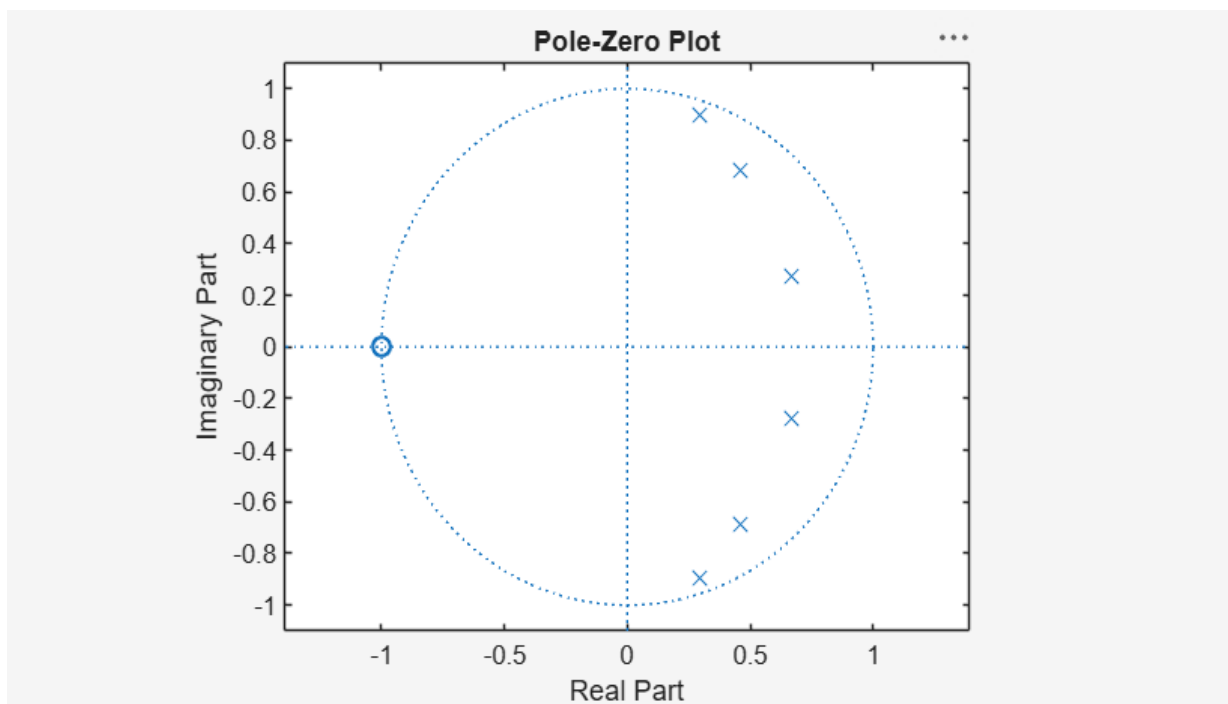


Fig 4:



EX NO:2	Chebyshev IIR Filter
DATE:	2.2. Design and Performance Evaluation of a Chebyshev Type-I IIR High-Pass Filter

Code:

```

clc;
clear;
close all;

%-----
% EX NO : 2
% TITLE : Design and Performance Evaluation of a
%         Chebyshev Type-I IIR High-Pass Filter
%         using MATLAB
%-----

% Filter Specifications
Rp = 1;      % Passband Ripple (dB)
Rs = 40;     % Stopband Attenuation (dB)
Wp = 0.50;   % Passband Edge Frequency (Normalized)
Ws = 0.35;   % Stopband Edge Frequency (Normalized)

% Determine Minimum Filter Order
[N, Wn] = cheb1ord(Wp, Ws, Rp, Rs);

% Design Chebyshev Type-I High-Pass Filter
[b, a] = cheby1(N, Rp, Wn, 'high');

%-----
% Check Stability
%-----
if all(abs(roots(a)) < 1)
    stability = 'STABLE';
else
    stability = 'UNSTABLE';
end

%-----
% Frequency Response
%-----
figure;
freqz(b, a, 1024);
title('Frequency Response of Chebyshev Type-I High-Pass Filter');

%-----

```

```

% Impulse Response
%-----
figure;
impz(b, a);
title('Impulse Response');

%-----
% Group Delay
%-----
figure;
grpdelay(b, a);
title('Group Delay');

%-----
% Pole-Zero Plot
%-----
figure;
zplane(b, a);
title('Pole-Zero Plot');

%-----
% Display Results
%-----
fprintf('-----\n');
fprintf('Chebyshev Type-I High-Pass Filter\n');
fprintf('-----\n');
fprintf('Filter Order (N) = %d\n', N);
fprintf('Normalized Cutoff Frequency (Wn) = %.4f\n', Wn);
fprintf('Passband Edge Frequency (Wp) = %.2f\n', Wp);
fprintf('Stopband Edge Frequency (Ws) = %.2f\n', Ws);
fprintf('Passband Ripple = %d dB\n', Rp);
fprintf('Stopband Attenuation = %d dB\n', Rs);
fprintf('System Stability = %s\n', stability);
fprintf('-----\n');

```

Output:

Fig 1:

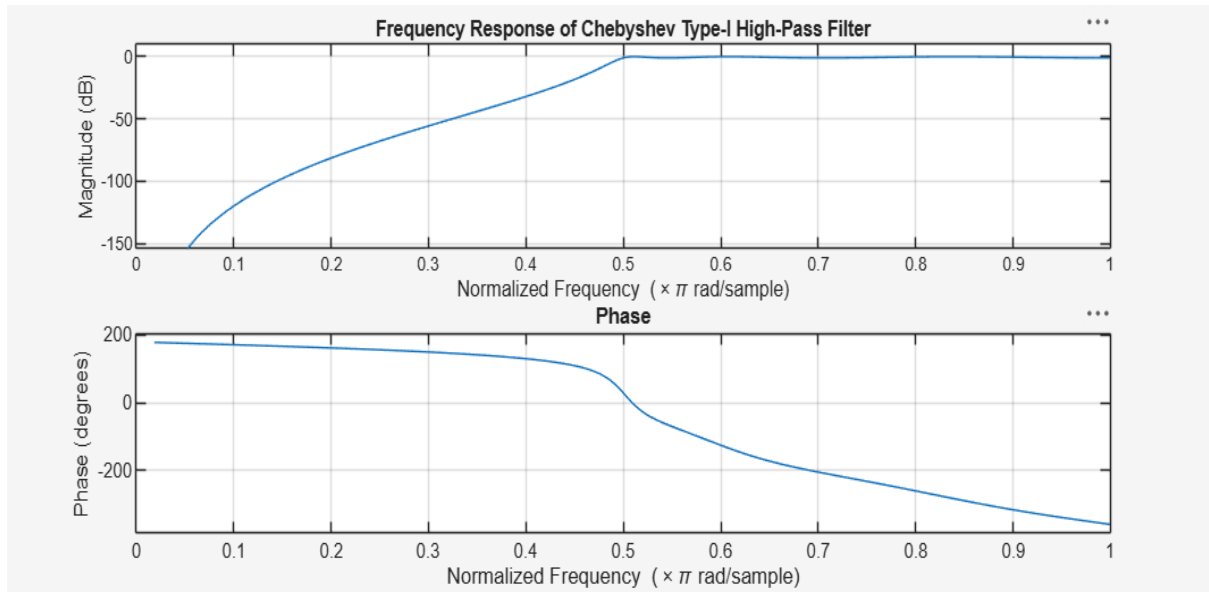


Fig 2:

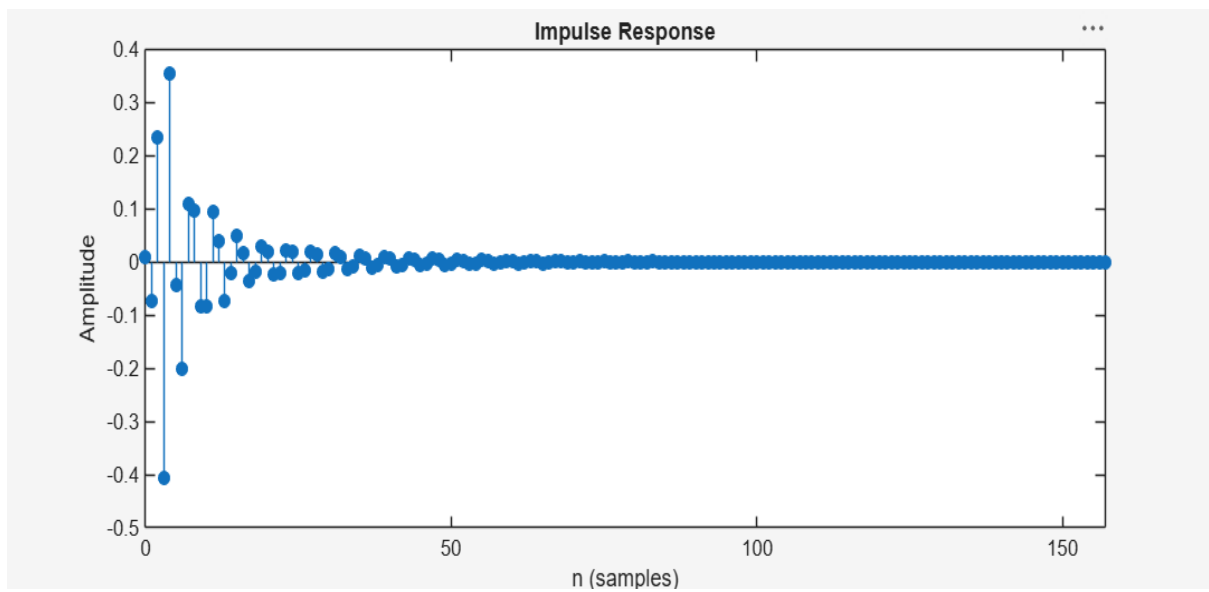


Fig 3:

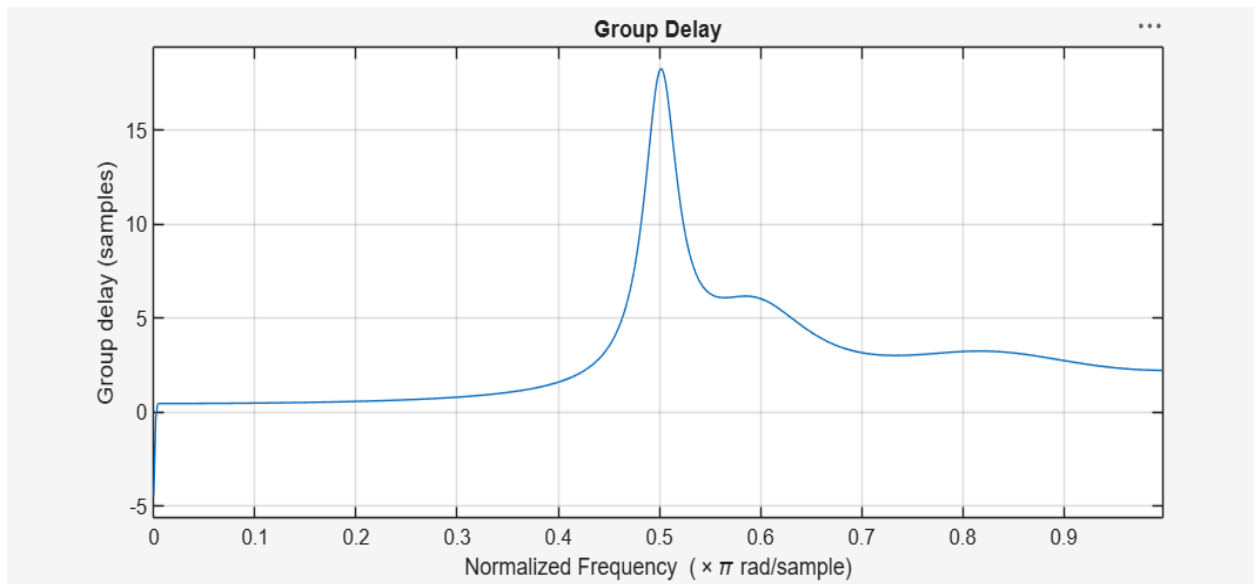


Fig 4:

